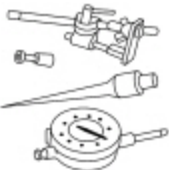
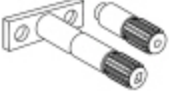
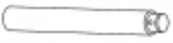


Differential Case and Ring Gear — Traction-Lok

 [Ford Ecat Electronic Parts Catalog](#)

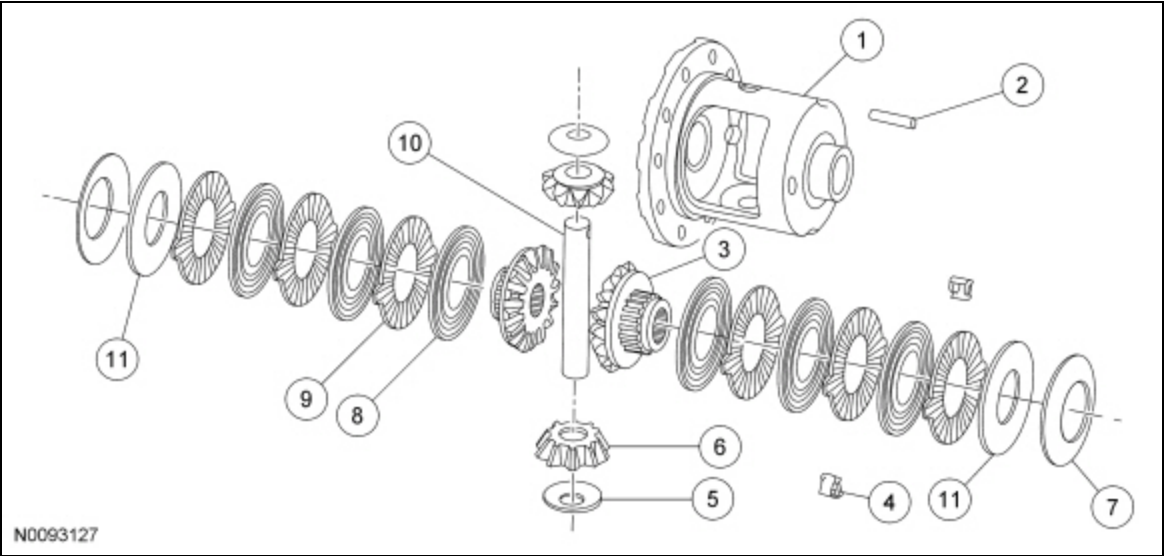
Special Tool(s)

 ST1214-A	Dial Indicator Gauge With Holding Fixture 100-002 (TOOL-4201-C)
 ST1374-A	Gauge, Differential Clutch 205-135 (T80P-4946-A)
 ST1265-A	Gauge, Differential Clutch 205-022 (T66L-4204-A)
 ST1372-A	Gauge, Differential Clutch (Mandrel for 205-135 [T80P-4946-A]) 205-270 (T87T-4946A)
 ST1749-A	Gauge, Differential (Traction-Lock) 205-386 (T97T-4205-D)
 ST1375-A	Installer, Differential Side Bearing 205-010 (T57L-4221-A2)
 ST1220-A	Remover, Differential Bearing 205-116 (T77F-4220-B1)

 ST1858-A	Handle 205-153 (T80T-4000-W)
 ST1543-A	Step Plate 205-D016 (D80L-630-5) or equivalent

Material

Item	Specification
Motorcraft® Additive Friction Modifier XL-3 (US); CXL-3 (Canada)	EST-M2C118-A
Motorcraft® SAE 75W-140 Synthetic Rear Axle Lubricant XY-75W140-QL (US); CXY-75W140-1L (Canada)	WSL-M2C192-A and GL-5
Motorcraft® Premium Long-Life Grease XG-1-E1	ESA-M1C75-B
Motorcraft® Threadlock and Sealer TA-25-B	WSK-M2G351-A5



Item	Part Number	Description
1	4204 4204	Differential case
2	4241 4241	Differential pinion shaft lock bolt
3	4236 4236	Differential side gear
4	—	Retainer clip (part of 4880)
5	4230 4230	Differential pinion thrust washer
6	4215 4215	Differential pinion gear
7	—	Plate preload spacer (Bellville spring) (part of 4880)

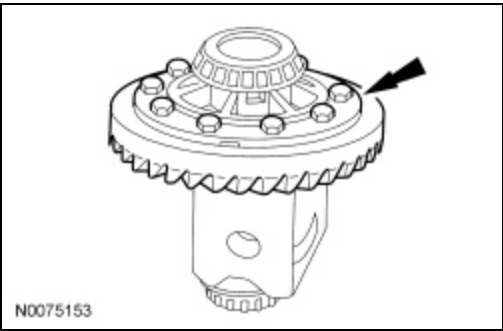
8	—	Clutch disc (part of 4880)
9	—	Clutch plate (part of 4880)
10	4211 4211	Differential pinion shaft
11	—	Selective shim (part of 4880)

Initial disassembly

NOTE: *If the Diagnostic Chart directed the servicing of the Traction-Lok clutch packs, it is not necessary to remove the differential carrier assembly from the axle housing. Proceed to Step 7.*

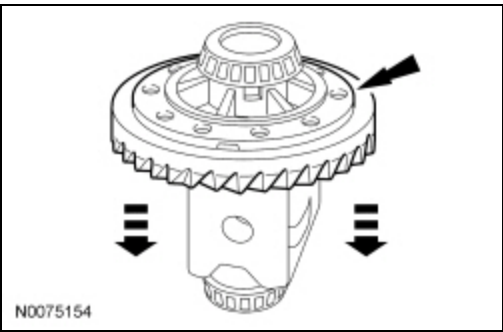
NOTE: *If the differential ring gear backlash between teeth is greater than specification, remove the ring gear to check the differential case flange runout. Do not disassemble the differential carrier assembly.*

1. Remove the differential carrier. Tag the differential shims and bearing cups for location. Refer to Differential Carrier.
2. Remove and discard the 10 differential ring gear bolts.

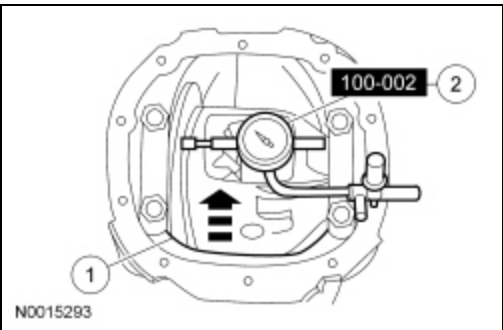


3. **NOTICE:** Care should be taken not to damage the differential ring gear bolt hole threads.

Insert a punch in the differential ring gear bolt holes and drive the differential ring gear off.

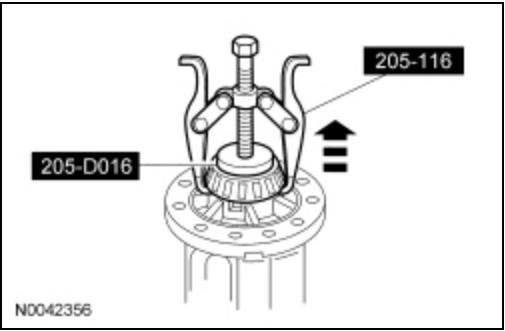


4. If checking differential case flange runout, install the bearing cups in their original location and install the differential assembly.
 1. Install the bearing caps and differential shims.
 - Rotate the differential assembly to verify the bearings have seated.
 2. Install the Dial Indicator Gauge With Holding Fixture and record the differential case runout.

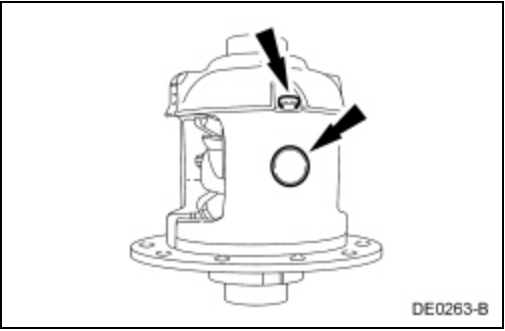


Final disassembly

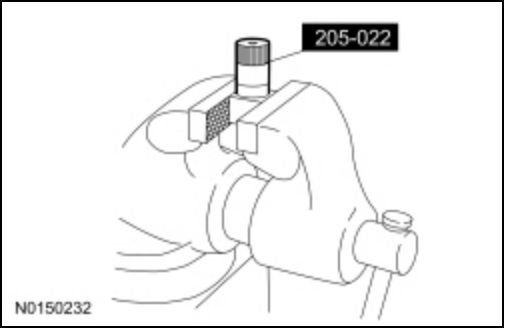
5. Using the Differential Bearing Remover and Step Plate, remove the differential bearings.



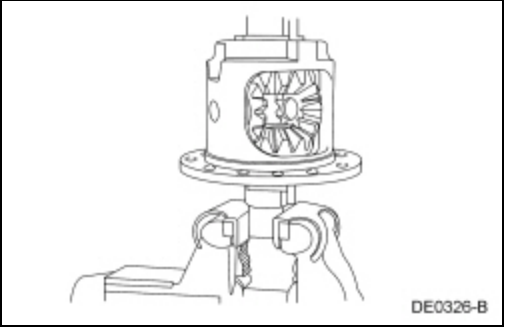
6. Remove the differential pinion shaft lock bolt and remove the differential pinion shaft.



7. Install the Differential Gauge in a suitable vise.

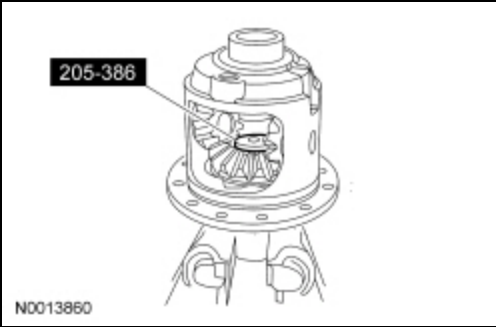


8. Install the differential case on the Differential Gauge.

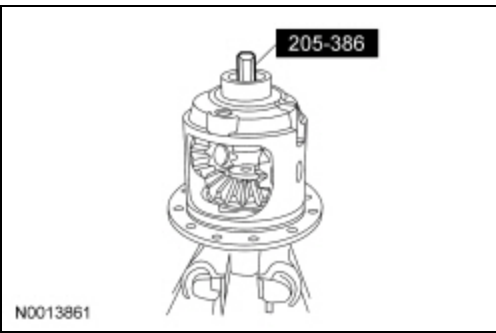


9. **NOTE:** Apply a small amount of grease to the centering hole of the Differential Gauge.

Install the Differential Gauge step plate in the bottom differential side gear.

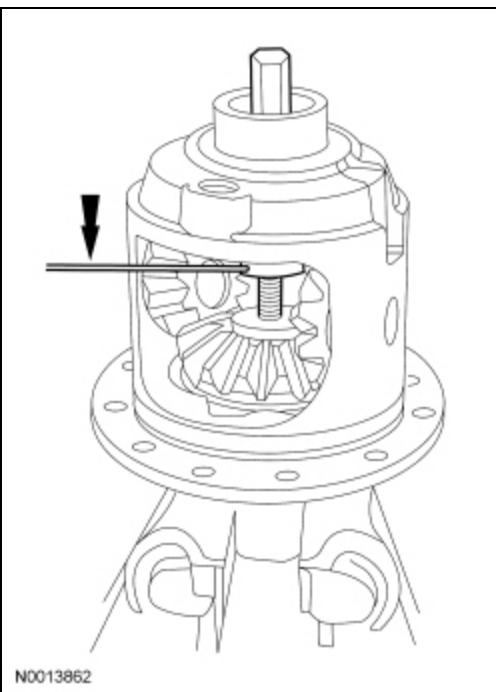


10. Install the nut in the upper differential side gear. Hold the nut in position while installing the hex screw. Tighten the hex-head screw until contact is made with the Differential Gauge step plate.



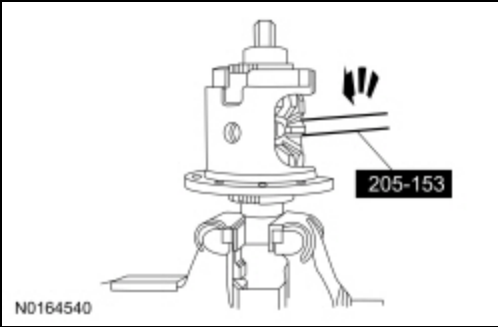
11. **NOTE:** *The dowel bar is used to keep the nut from turning when the forcing screw is tightened.*

Insert a suitable dowel bar in the hole of the nut. Tighten the forcing screw to force the differential side gears away from the differential pinion gears.

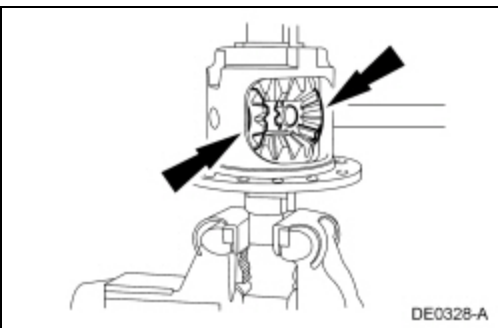


12. **NOTE:** *Differential pinion gear thrust washers cannot be removed independently of the differential pinion gears.*

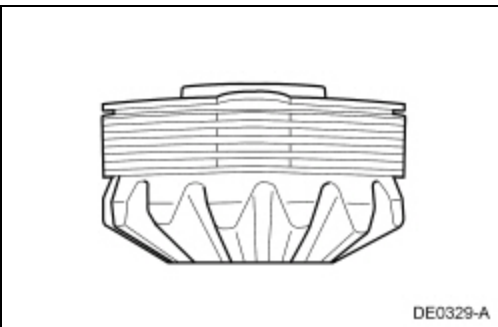
Insert the Differential Rotator in the pinion shaft bore, and turn the differential case to "walk" the differential pinion gears and differential case windows.



13. Remove the differential pinion gears and differential pinion gear thrust washers.



14. Remove the differential side gears and differential clutch packs with the selective shim and Belleville spring, tag them RH and LH.



15. **NOTICE: Do not use acids or solvents when cleaning the differential clutch pack or damage to the components can result. Wipe components with a clean, lint-free cloth only.**

NOTE: *When separating the clutch plates and clutch disc, note the sequence in which they are disassembled. They must be reassembled in the same sequence.*

Separate the differential clutch disc and clutch plates for cleaning and inspection.

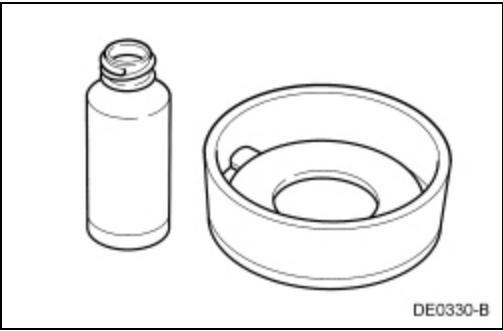


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Assembly

1. **NOTE:** Use 118 ml (4 oz) of the specified friction modifier in the axle.

Pre-lubricate each steel clutch plate and soak all friction plates with friction modifier for at least 15 minutes.

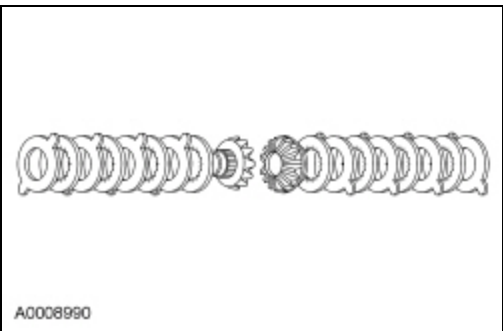


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2. **NOTE:** Do not mix the differential clutch packs or selective shims from one side with the other.

NOTE: The Belleville spring is a dished plate.

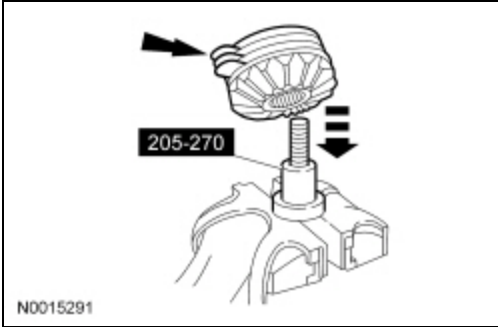
Assemble the differential clutch packs (without the selective shims and Belleville spring) on the respective differential side gears.



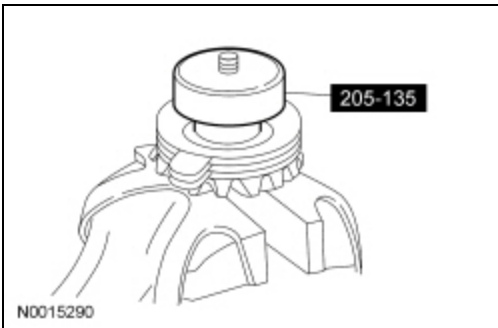
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3. **NOTE:** Make sure the correct mandrel is used with the Differential Clutch Gauge.

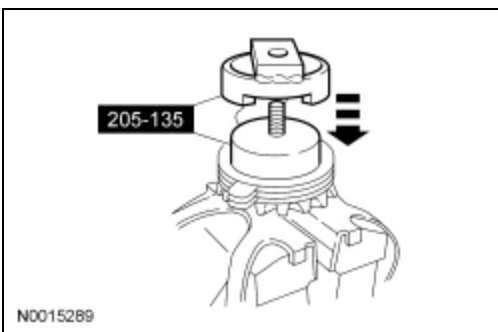
Place the base portion of the Differential Clutch Gauge in a vise. Install the differential clutch pack and differential side gear (without the selective shim or Belleville spring) on the gauge.



4. Position the Differential Clutch Gauge hand-tight on top of the differential clutch pack.

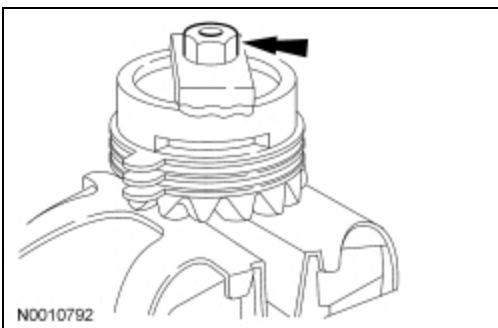


5. Install the Differential Clutch Gauge over the disc and differential clutch pack.



6. Install the nut of the gauge over the top and base stud.

- Tighten to 7 Nm (62 lb-in).



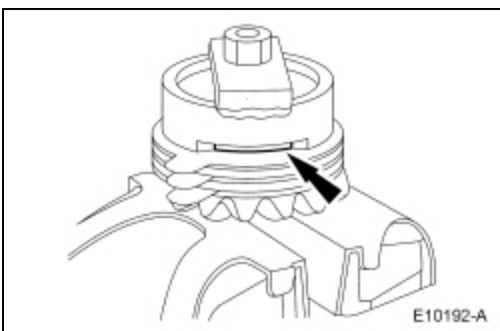
7. **NOTE:** Selective shims shown are available as part of the Clutch Pack Replacement Kit.

NOTE: The belleville spring thickness (0.055 in.) and the selective shim should be the same as the feeler gauge measurement. Do not use more than 2 selective shims.

Use the feeler gauge and select the thickest blade that will enter between the tool and the differential clutch pack. Refer to the chart to determine which shim to use.

Selective Shims

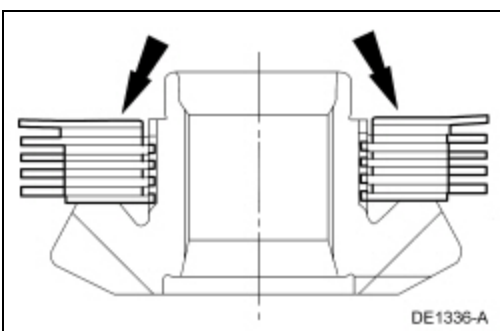
Feeler Gauge	Description
0.072-0.080	0.025 in
0.081-0.085	0.030 in
0.086-0.090	0.035 in
0.091-0.095	0.040 in
0.096-0.101	0.045 in
0.102-0.107	0.050 in
0.108-0.113	0.055 in



8. Remove the Differential Clutch Gauge from the clutch pack and differential side gear assembly.

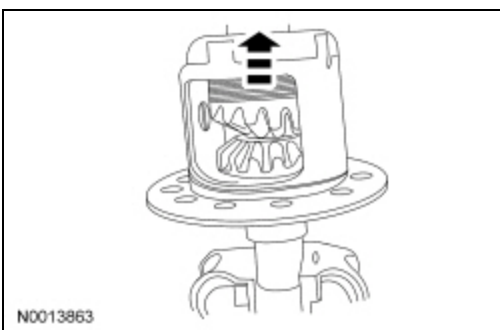
9. Place the selective shim and Belleville spring on the differential clutch pack.

- The dished or concave side of the Belleville spring must face up against the thrust face of the differential case.



10. Install the differential side gears with the differential clutch packs, selective shims and Belleville spring into the differential case.

- Hold the differential side gear assembly in place to prevent it from falling out of the differential case.

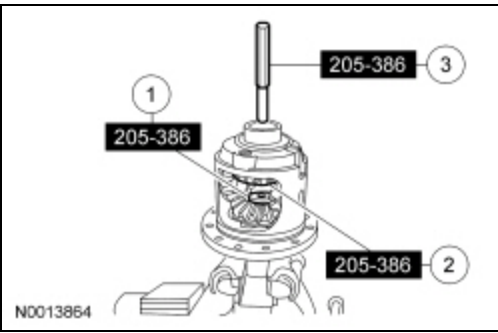


11. **NOTE:** Apply a small amount of grease to the step plate bore.

NOTE: If necessary, insert the dowel bar in the nut bore to keep the nut from turning as the hex screw is tightened.

Assemble the Differential Gauge to the differential case.

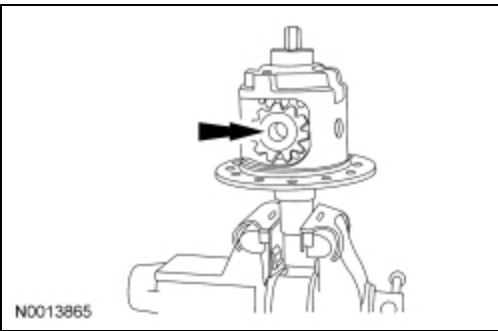
1. Position the Differential Gauge step plate in the bottom differential side gear.
2. Position the Differential Gauge in the top differential side gear and hold it in place.
3. Install the Differential Gauge and tighten it 2 turns after it contacts the bottom step plate.



12. **NOTE:** *Lubricate both sides of the differential pinion gear thrust washers with axle lubricant.*

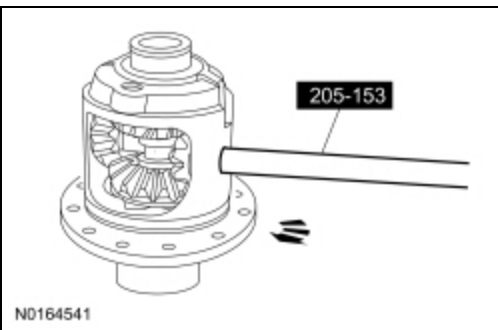
NOTE: *Make sure the differential pinion gears are 180 degrees apart so they will align correctly with the pinion shaft bore.*

Position the differential pinion gears and differential pinion gear thrust washers in the window of the differential case so they mesh with the differential side gear teeth.



13. **NOTE:** *It will probably be necessary to loosen or tighten the forcing screw to allow the differential pinion gears and differential side gears to rotate.*

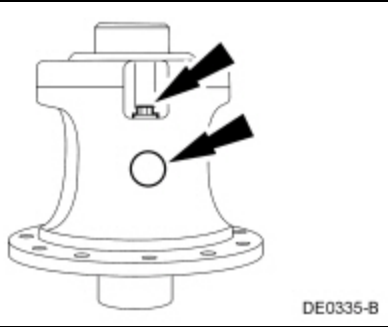
Inspect the Differential Rotator into the pinion shaft bore, and turn the differential case. This will cause the differential pinion gears to engage the differential side gear and "walk" into the differential case. Rotate the differential case until the pinion shaft mating holes are lined up exactly with the holes in the differential pinion gears.



14. Remove the special tools.

15. **NOTE:** *If a new differential pinion shaft lock bolt is unavailable, coat the threads of the original differential pinion shaft lock bolt with threadlock prior to installation.*

Install the differential pinion shaft, then a new differential pinion shaft lock bolt finger-tight.

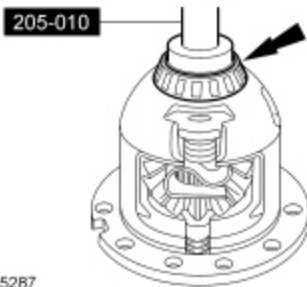


16. Check the torque required to rotate one differential side gear.

- Mount the differential assembly and Differential Clutch Gauge in a vise.
- The initial minimum break-away torque, if original clutch plates are used, must be within specification. The minimum rotating torque required to keep the differential side gear turning with new clutch plates may vary.



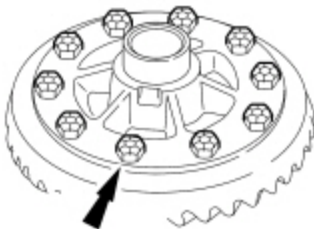
17. Using the Differential Side Bearing Installer, install the 2 differential bearings on the differential case.



18. Position the differential ring gear and the differential case. Align the bolt holes by hand-starting 2 new bolts through the holes in the differential case and into the differential ring gear.

19. Hand-start the 8 remaining new differential ring gear bolts. Tighten the 10 bolts in 2 stages.

- Stage 1: Tighten to 60 Nm (44 lb-ft).
- Stage 2: Tighten an additional 90 degrees.



20. Install the differential carrier. Refer to Differential Carrier.